

Systems that refuse to invent.

Kunal Kumar

Computer Science, class of 2028 · Bengaluru, India
Frontend Engineering Intern, Emergent (YC S24)

MATS round 2

Activate AI Fellow · 1 of 15 from 500+

HPAIR 2026, Harvard

Hack Club · 1,500 students reached

50+ students taught

29 patches merged upstream

Every system I use will guess, then deliver the guess in the same voice it uses for facts. That is not a bug. An honest “I don't know” is harder to build than a confident answer, and nobody is rewarded for building it.

So that is what I build. Five projects, one idea, arrived at five separate times before I noticed it had become a thesis.

The sixth page is the rest of me: two years of running events, teaching, and shipping code to people who owed me nothing.

32 repositories. 829 contributions in twelve months. 29 patches merged into 35 open-source projects, approved by 10 maintainers.

01	02	03	04	05
Recall	ATTEST	Support Triage	Minutes	PDFChat
A memory engine for unfinished work	Money that moves only on proof	An agent that cannot be talked into it	A meeting workspace, rebuilt	Cites its pages, or refuses

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

Recall

0

MODELS IN THE PRODUCTION PATH

A memory engine for unfinished work. One keystroke brings back the tabs, files and chats from your last session.

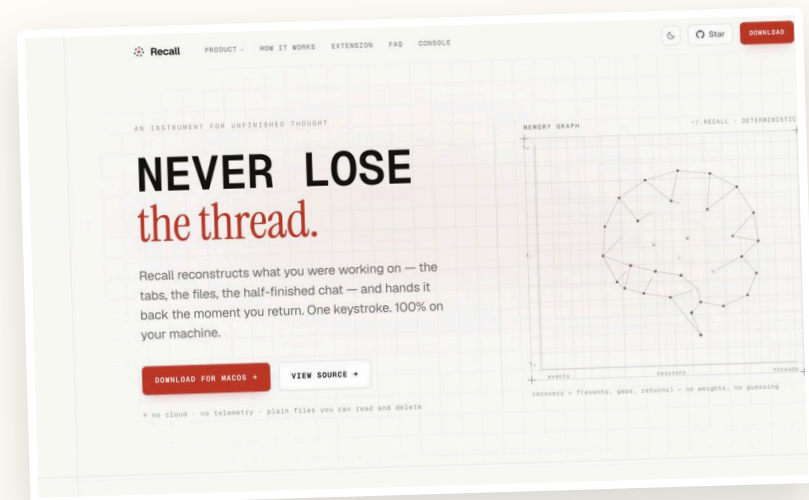
Someone interrupts you at 3pm. You sit back down and everything is still there, the file, the tab, the branch, and none of it helps. **You did not lose files. You lost continuity.**

Spotlight and `grep` assume you remember a string. A cognitive restart never asks that. It asks *what was I doing*, and nothing on the machine answers that question.

Press **Ctrl + Space**, type half a thought, and the moment, the session and the topic all come back at once. An empty query offers what to resume, **capped at four items**, because showing you everything relevant is the same as showing you nothing.

TECHNICAL ASPECTS

Seven layers over a local event log, fed by a Chrome extension and a file watcher. **No model sits in the production path**, so the same question can never return two different pasts. A 35-section smoke test enforces it. No cloud, no account, no telemetry: everything Recall knows about you sits in `~/.recall/` as plain JSON, readable with `cat`. It ships as a 26 MB menu-bar app that reports degraded features instead of hiding them.



YEAR	2026
ROLE	Sole author. Engine, launcher, extension, daemon, docs.
BUILT WITH	Python · PyQt6 · FastAPI · ChromaDB · Tauri
LINK	web-seven-puce-99.vercel.app github.com/kunalKumar-13/Recall-me

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

ATTEST

₹0.00

MOVED AGAINST THE WRONG ACCOUNT

A financial control layer. It verifies what an AI proposes before anything is allowed to move money.

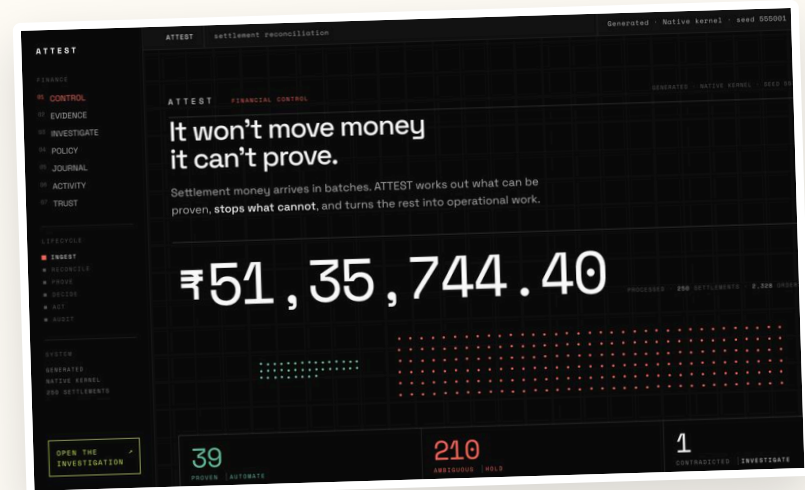
A payment gateway sends one lump sum covering many orders. Somebody has to work out what it was made of, and **more than one set of orders can add up to the same credit exactly.**

Those sets discharge *different customers*. Guessing marks the wrong customer as paid while the books still balance, so no alarm fires and nobody finds out until it is expensive.

So ATTEST refuses. Across 500 settlements it had never seen, **82.4% were correctly refused and ₹0.00 moved against the wrong account.** The refusal is the product, not a failure rate.

TECHNICAL ASPECTS

Four stages, and the separation between them is the design. An **advisor** proposes which records to look at, and is **never given an amount**. A **verifier** sharing no code with it re-derives every explanation independently. A **policy** automates only where measured error costs less than a human check. Integer paise end to end, so no float ever touches money. The published benchmark names a competitor that beats mine on precision, because a comparison curated in your own favour proves nothing.



YEAR	2026
ROLE	Sole author. Architecture, solver, evaluation harness.
BUILT WITH	Python · Rust solver kernel
LINK	github.com/kunalKumar-13/attest

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

Support Triage

132/132

EVALUATION TESTS PASSING

A support agent where code owns the control flow and the model is a constrained analyst, not the decision-maker.

A support agent that can be talked into a destructive action is worse than no agent at all. The usual design hands the model the wheel and hopes the prompt holds.

This one does not. **The LLM is invoked exactly once per ticket, inside an unforgeable sentinel block**, and returns a typed JSON object that a policy validator adjudicates. It cannot decide escalation, cannot execute tools, and never sees raw PII.

Its worst possible behaviour is *escalate to a human*. **132 of 132 evaluation tests pass**, and two runs on the same input produce a byte-identical file.

TECHNICAL ASPECTS

Untrusted ticket text never leaves its frame, and retrieved chunks are grounding rather than instructions. It defends against prompt injection in seven languages, plus zero-width characters, homoglyphs and base64 obfuscation. When retrieved chunks contradict each other, **confidence drops and the validator escalates** rather than picking a side. Every tool call is schema-validated, risk-gated and idempotency-checked.

```
$ triage --ticket 4417

  ingest  ticket text sealed in <<<TICKET>>>
 retrieve 4 chunks · 2 disagree numerically
 analyse  LLM called once, inside sentinel block
          → typed JSON, schema validated
 policy    consensus broken → confidence 0.41
          escalation cap 0.60 · not eligible

DECISION  ESCALATE TO HUMAN
reason     retrieval_conflict, low_confidence

$ sha256sum run_a.csv run_b.csv
2f9c...a71b run_a.csv
```

YEAR	2026
ROLE	Sole author. Pipeline, policy validator, threat model.
BUILT WITH	Python · LlamaIndex · BM25 · TF-IDF
LINK	github.com/kunalKumar-13/ deterministic-support-triage

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

Minutes

2

COMMANDS TO A POPULATED PRODUCT

A meeting workspace with interactive transcripts, AI notes and workspace-wide search, rebuilt from the product outward.

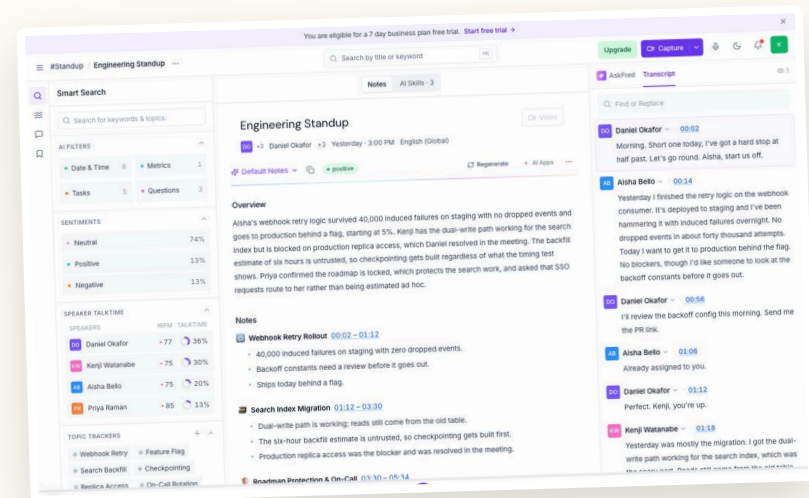
“Clone this product” is the easiest brief in the world to do badly. You rebuild the hero section and ship something that photographs well and collapses when anyone clicks.

But a meeting product is not its landing page. It is what happens on the fourth day, when you half-remember someone committing to something about pricing and you need to find it. **Everything valuable happens after the meeting ends.**

So I set a bar I could not fake: everything except the speech-to-text actually works, and **the README says so on the first screen**, before a reviewer can discover it.

TECHNICAL ASPECTS

Next.js 15 over FastAPI, with **SQLite FTS5** as a real full-text index rather than a LIKE query in a costume. Auth is enforced at the edge, so a signed-out visitor never receives app UI at all. First boot seeds seven complete meetings, which makes `git clone` to a populated product a two-command job. The decision that mattered: I stopped matching screens and **extracted the token system instead**, after which twenty-plus screens came out coherent untouched.



YEAR	2026
ROLE	Sole author. Frontend, backend, schema, search, summariser. Built to an assignment brief.
BUILT WITH	Next.js 15 · React 19 · TypeScript · FastAPI · SQLite
LINK	github.com/kunalKumar-13/minutes

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

PDFChat

0

UNCITED ANSWERS

Ask a document anything. Every answer cites its pages, or says plainly that the document does not support the question.

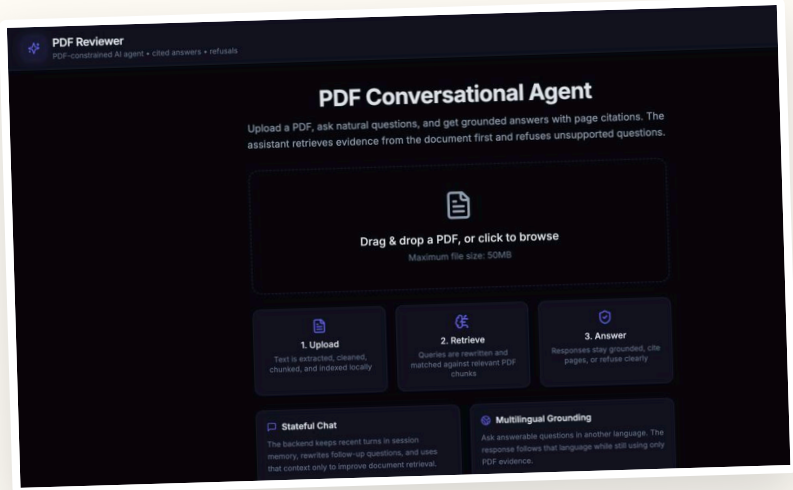
A confident wrong answer costs more than no answer. Retrieval gets messy, and the standard failure is a model that fills the gap fluently and invisibly.

PDFChat will not. Every response either **cites the pages it came from** or refuses clearly. Multi-turn chat keeps session context for retrieval only, so the document stays the sole evidence.

It answers in your language while still treating the source as the only truth, which is exactly the case where grounding usually breaks quietly.

TECHNICAL ASPECTS

Extraction, chunking and indexing all run locally, and queries are rewritten against the relevant chunks before generation. The refusal path is a first-class branch rather than an error handler: if retrieval does not support an answer, that *is* the answer. It handles documents up to 50 MB, and it is deployed and publicly usable right now.



YEAR	2026
ROLE	Sole author. Retrieval, citation and refusal path.
BUILT WITH	Python · PyMuPDF · TF-IDF retrieval · FastAPI
LINK	pdf-reviewer-ten.vercel.app github.com/kunalKumar-13/PDF-Reviewer

NAME	SCHOOL	CONTACT	PORTFOLIO SUBMISSION FOR
Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

● BEYOND THE FIVE PROJECTS

The rest of me

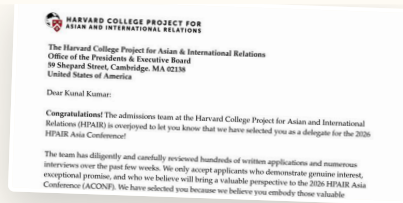
Research, rooms I did not belong in, people who owed me nothing, and 1,500 students.



I want to know what a system is allowed to cause

Through to **round 2 at MATS**, and selected for the **Activate AI Fellowship**, India's first cohort.

Three projects here turned out to be about one question before I had a name for it: when a model proposes something, what is *structurally* permitted to believe it? **I read alignment work the way other people read the news.**



I get more from rooms I do not belong in

Selected delegate to **HPAIR 2026 at Harvard**, arguing about things I had no technical claim on, with people from unrelated fields.

Every project here began as a problem I could name in one sentence. **That habit came from that room, not from a codebase.**



Strangers grade my code, and I keep going back

Merged into OpenStreetMap's editor, Django's own site, a Bitcoin Lightning wallet, MetaBrainz, Plone and the **Fortran standard library**, in four languages.

I had no reason to know Fortran. Landing a patch meant learning how those scientists argue about correctness *before* writing a line. **That is the skill: arriving somewhere I have no standing and becoming useful.**



I make things happen for other people

Growth & Ops Lead at **Hack Club**: infrastructure and logistics for Campfire, Daydream and CodeDay Bengaluru, reaching **1,500 students**. Now Ops Lead, Open Source Club; Head of Immersions, Aspirants Relations.

At that scale every failure is one you did not plan for, so you design the degradation instead. **Every system here has a documented failure mode, and that came from a wifi outage.**

Why this Academy

Everything I build lives on a screen someone is already sitting at. I have never built for the device someone holds *while something is happening to them*, and that is a different discipline, not the same one scaled down. Three times now the hardest part of an engineering project turned out to be a design problem I had no vocabulary for: ranking in Recall was an editorial question rather than a mathematical one, and the hardest thing in ATTEST is not the solver, it is making an operator understand at a glance why the machine held the money. **An honest "I don't know" is worth nothing if nobody can read it.** That is the half I am asking you to teach me.

NAME

Kunal Kumar

SCHOOL

Computer Science · Student

Class of 2028

CONTACT

kunalsain0324@gmail.com

+91 85448 88032

linkedin.com/in/sainkunal ·

github.com/kunalKumar-13

PORTFOLIO SUBMISSION FOR

Apple Developer Academy Indonesia

@ BINUS, Bali · Cohort 2027 · Tech track